

D683

Advanced AI and ML

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An online furniture retailer with three warehouses wants to predict how many days each order will take to arrive. It has history on 400,000 orders, and the team prefers Python. Which problem type fits?

- A. Clustering.
- B. Regression.
- C. Binary classification: late or on time.
- D. Whatever type the team's favorite Python library handles best.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Before building a deep neural network to predict weekly store sales, the ML lead asks for a baseline. The team just hired a data scientist with a PhD. What is the best baseline?

- A. Any baseline that reaches 90% accuracy.
- B. Skip it; the new PhD will know if the network is good.
- C. Build the deep network first and compare later versions to it.
- D. Last year's sales for the same week, or a simple linear regression.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A churn model trained on eight years of records includes a feature called "account closure reason code," which staff fill in only after a customer cancels. It scores 0.97 AUC in testing but performs far worse in production. What is the problem?

- A. Too little history; add more years of data.
- B. Target leakage; remove features not available at prediction time.
- C. A high AUC always means leakage.
- D. Overfitting; add regularization.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A model predicts, at the moment an insurance claim is filed, whether it is fraudulent. Which feature is safe to use?

- A.** The claimant's number of prior claims before this one.
- B.** The payout amount approved weeks later.
- C.** The claim's ID number.
- D.** The investigator's final fraud finding.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A k-nearest neighbors model uses annual income (from \$20,000 to \$400,000) and number of support calls (0 to 12). The model seems to ignore support calls. The team runs it on a laptop with 16 GB of memory. Why?

- A.** The laptop lacks memory.
- B.** k is too small.
- C.** Income's scale dominates the distance; standardize both features.
- D.** Support calls have no predictive value.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A model screens mammogram images to flag cases for radiologist review. Missing a cancer is far more costly than an extra review. The images come from a partnership with four clinics. Which metric should the team prioritize?

- A.** Number of partner clinics.
- B.** Precision.
- C.** Accuracy.
- D.** Recall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which use case should prioritize precision over recall?

- A. Automatically locking customer accounts for suspected fraud, where each false flag locks out a good customer.
- B. Screening for a serious disease.
- C. Every use case should weigh them equally.
- D. Screening luggage for weapons.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A classifier has precision 0.60 and recall 0.90. It retrains every Sunday on a cloud GPU. What is its F1 score?

- A. 0.72
- B. 0.75
- C. 0.90
- D. 0.54

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A classifier has $F1 = 0.80$ and precision = 1.00. What is its recall?

- A. 0.60
- B. 0.80
- C. 0.20
- D. About 0.67

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Six months after launch, a loan default model's precision falls from 0.82 to 0.61. The code has not changed, and the app was redesigned with a new logo. Most applicants now come from a new online partner channel. What is the best explanation and action?

- A. Validated models never need retraining.
- B. The new logo confused applicants.
- C. Data drift; compare new inputs to training data, then retrain on recent data.
- D. A hidden code bug.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A team runs 5-fold cross-validation on 10,000 rows, using random seed 42. In each fold, how many rows are used for training and how many for validation?

- A.** 10,000 for both.
- B.** 8,000 training, 2,000 validation.
- C.** 2,000 training, 8,000 validation.
- D.** 9,000 training, 1,000 validation.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
-----------------------------	---	-----------------------	---

12. [TRANSFER — reversed stem / different variable]

Four-fold cross-validation gives scores of 0.80, 0.84, 0.78, and 0.82. What score should be reported?

- A.** 3.24
- B.** 0.84
- C.** 0.81
- D.** 0.80

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
-----------------------------	---	-----------------------	---

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

An online furniture retailer ~~with three warehouses~~ wants to predict how many days each order will take to arrive. It has history on 400,000 orders, ~~and the team prefers Python~~. Which problem type fits?

Target word: problem framing **Decides:** The target is a number of days.

A	Confusion set	Finds groups without a target; there is a target here.
B	Correct	Predicting a continuous or count value.
C	Adjacent category	Throws away the number of days that was asked for.
D	Camouflage	Tool preference does not decide framing.

2. Answer D Concept 2

Before building a deep neural network to predict weekly store sales, the ML lead asks for a baseline. ~~The team just hired a data scientist with a PhD~~. What is the best baseline?

Target word: baseline choice **Decides:** A baseline must be simple and cheap to compare against.

A	Absolute language	A baseline is simple by design; it has no required score.
B	Camouflage	Credentials do not replace a comparison point.
C	Inverted operation	Reverses the order; the simple model sets the bar.
D	Correct	Shows whether the complex model earns its cost.

3. Answer B Concept 3

A churn model ~~trained on eight years of records~~ includes a feature called “account closure reason code,” which staff fill in only after a customer cancels. It scores 0.97 AUC in testing but performs far worse in production. What is the problem?

Target word: diagnose the gap **Decides:** The feature exists only after the outcome.

A	Camouflage	History length does not decide the leak.
B	Correct	The code reveals the answer in training but is blank for live customers.
C	Absolute language	High scores can be real; here the feature timing is the proof.
D	Adjacent category	Regularization will not remove a leaked feature.

4. Answer A Concept 3 · transfer

A model predicts, at the moment an insurance claim is filed, whether it is fraudulent. Which feature is safe to use?

Target word: safe feature — reversed **Decides:** Must be known at filing time.

A	Correct	Known when the claim is filed.
B	Adjacent category	Created after the prediction point.
C	Confusion set	Available but meaningless; an identifier can only add noise or leak order.
D	Inverted operation	That is the label itself.

5. Answer C Concept 4

A k-nearest neighbors model uses annual income (from \$20,000 to \$400,000) and number of support calls (0 to 12). The model seems to ignore support calls. ~~The team runs it on a laptop with 16 GB of memory.~~ Why?

Target word: why a feature is ignored **Decides:** Income's range dwarfs the call count in distance.

A	Camouflage	Memory does not decide distance math.
B	Confusion set	k changes neighbor count, not feature weight.
C	Correct	Scaling puts features on equal footing.
D	Dropped element	Jumps to a conclusion without fixing scale.

6. Answer D Concept 5

A model screens mammogram images to flag cases for radiologist review. Missing a cancer is far more costly than an extra review. ~~The images come from a partnership with four clinics.~~ Which metric should the team prioritize?

Target word: metric choice **Decides:** False negatives are the costly error.

A	Camouflage	Not a model metric.
B	Inverted operation	Precision limits false alarms, the cheaper error here.
C	Adjacent category	Hides misses when cancer is rare.
D	Correct	Recall measures how many true cases are caught.

7. Answer A Concept 5 • transfer

Which use case should prioritize precision over recall?

Target word: use case — reversed **Decides:** False positives must be the costly error.

A	Correct	False alarms are the costly error.
B	Inverted operation	Misses are costly; recall comes first.
C	Absolute language	The cost of each error differs by case.
D	Confusion set	A miss is far worse than an extra check.

8. Answer A Concept 6

A classifier has precision 0.60 and recall 0.90. ~~It retrain every Sunday on a cloud GPU.~~ What is its F1 score?

Target word: F1 **Decides:** Harmonic mean of 0.60 and 0.90.

A	Correct	$2 \times 0.60 \times 0.90 / 1.50 = 1.08 / 1.50 = 0.72$.
B	Adjacent category	The simple average, not the harmonic mean.
C	Wrong stem number	Recall alone.
D	Dropped element	The product without the 2 / (P + R) step.

9. Answer D Concept 6 · transfer

A classifier has $F1 = 0.80$ and precision = 1.00. What is its recall?

Target word: solve for recall **Decides:** $0.80 = 2R / (1 + R)$.

A	Adjacent category	Solves $(1 + R) / 2 = 0.80$, the simple average.
B	Wrong stem number	The F1 value.
C	Inverted operation	$1.00 - 0.80$ subtracts instead of solving.
D	Correct	$0.80 + 0.80R = 2R$, so $R = 0.80 / 1.20 \approx 0.667$.

10. Answer C Concept 7

Six months after launch, a loan default model's precision falls from 0.82 to 0.61. The code has not changed, ~~and the app was redesigned with a new logo.~~ Most applicants now come from a new online partner channel. What is the best explanation and action?

Target word: diagnose decline **Decides:** The applicant population changed.

A	Absolute language	Data and relationships change over time.
B	Camouflage	Branding does not decide model precision.
C	Correct	A new channel shifts the input distribution.
D	Dropped element	Drops the stated fact that the code did not change.

11. Answer B Concept 8

A team runs 5-fold cross-validation on 10,000 rows, ~~using random seed 42.~~ In each fold, how many rows are used for training and how many for validation?

Target word: fold sizes **Decides:** Each fold holds $10,000 / 5 = 2,000$ rows.

A	Dropped element	Drops the hold-out entirely.
B	Correct	Four folds train, one validates.
C	Inverted operation	Swaps the roles.
D	Confusion set	That is 10-fold.

12. Answer C Concept 8 · transfer

Four-fold cross-validation gives scores of 0.80, 0.84, 0.78, and 0.82. What score should be reported?

Target word: summarize folds **Decides:** Report the mean.

A	Inverted operation	The sum, not divided by the fold count.
B	Adjacent category	The best fold, which overstates performance.
C	Correct	$3.24 / 4 = 0.81$.
D	Dropped element	Only the first fold.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Framing the ML problem	Predicting a number is regression; predicting a category is classification; finding groups without labels is clustering.	Framing a numeric target as a category, or choosing by tool preference.	“How many,” “how much,” or “which group.”
2	Baseline first	Build a simple baseline before a complex model so you can tell whether complexity adds value.	Starting with the most advanced model and skipping comparison.	“Before building a deep network...”
3	Target leakage	Use only features that would exist at prediction time; features created after the outcome leak the answer.	Treating suspiciously high test scores as success.	A feature filled in after the event being predicted.
4	Feature scaling	Distance- and gradient-based models (k-NN, SVM, neural nets) need features on comparable scales.	Concluding a small-scale feature is useless.	Dollars next to small counts in a k-NN or SVM.
5	Metric follows error cost	Prioritize recall when missing a positive is costly; prioritize precision when a false alarm is costly.	Defaulting to accuracy.	“Missing a case is far worse than...”
6	F1 score	$F1 = 2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$, the harmonic mean.	Using the simple average.	A precision and a recall and “F1.”
7	Drift and monitoring	When input data or relationships shift after deployment, performance drops; monitor, investigate, and retrain.	Assuming a validated model stays valid.	Performance falls months later with no code change.
8	k-fold cross-validation	Split data into k folds; each fold validates once while the other k – 1 train; report the mean score.	Reporting the best fold or reversing train and validate sizes.	“5-fold” or fold scores listed.

Course scope (public description): Advanced AI and ML provides an opportunity for students to exercise their knowledge and skills in the design and development of artificial intelligence (AI) and machine learning (ML) solutions for real-world business problems. Through a hands-on project, students delve into the design and execution planning stages. The course culminates with the development of a fully functional AI/ ML product.